Module 02: Agents in Robotics

Learning Objectives

1. Define **Agents** within the context of Robotics.
2. Analyze how Agents interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Agents.

Introduction

This module explores **Agents**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Agents is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Agents as a Markov Blanket Boundary

How does Agents define the boundary between the agent and the environment?

2. Generative Models of Agents

What parameters involved in Agents must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Agents drive the perception-action loop?

Applications

In Robotics, we see Agents manifest in: * **Specific Example 1**: [Add domain-specific example here] * **Specific Example 2**: [Add domain-specific example here]

Conclusion

Understanding Agents allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.